

From Myelopathy to Myopathy

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Session Objectives

Objective 1- **Explain** how to localize based on clinical presentation.

Objective 2- **Describe** how to exclude treatable causes of potentially fatal disorders.

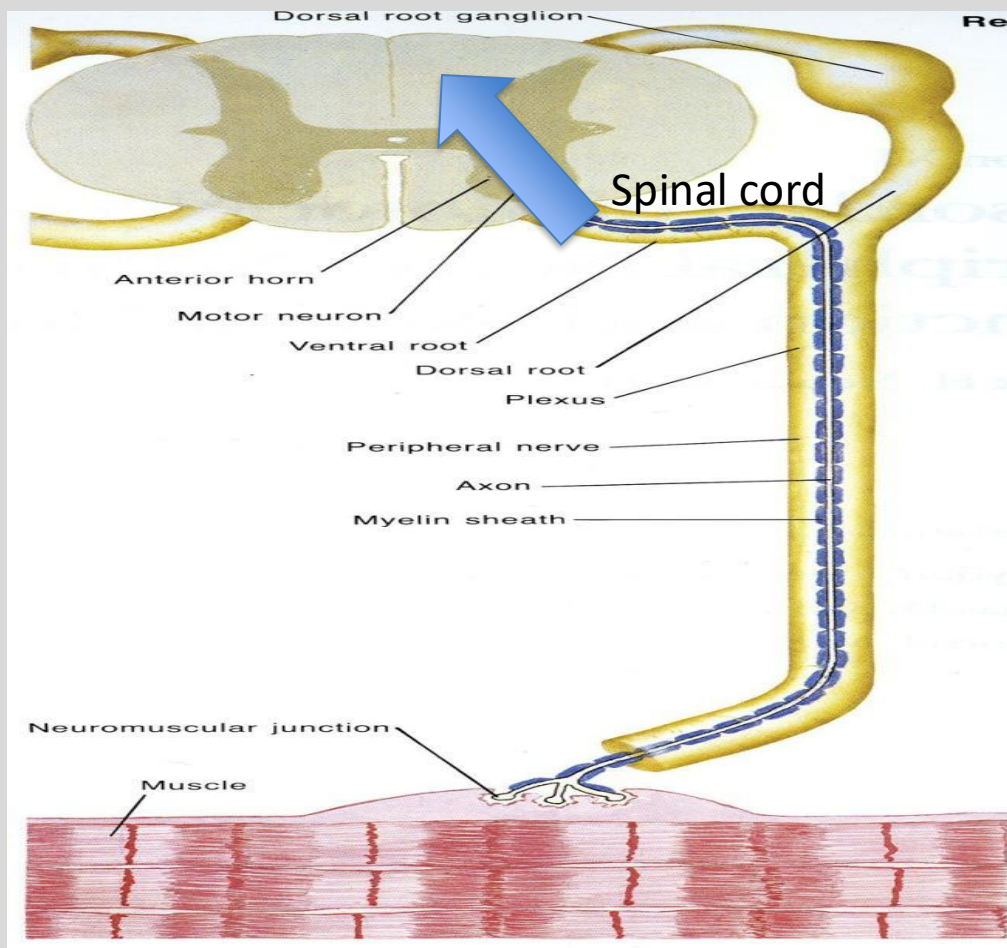
Objective 3- **Correlate** the differential diagnosis of disease at each level of the motor unit.

Objective 4- **Apply** treatment options, if there are any available.

Objective 5- **Identify** a neurological emergency

Overview

- Spinal Cord disorders (myelopathies)
- Anterior horn cell disorders
- Motor Neuron Disorders
- Ventral Root disorders (radiculopathies)
- Dorsal Root ganglion disorders
- Plexopathies
- Peripheral Neuropathies
- Demyelinating disorders and axonal changes
- Neuromuscular junction disorders
- Myopathies



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Case 1

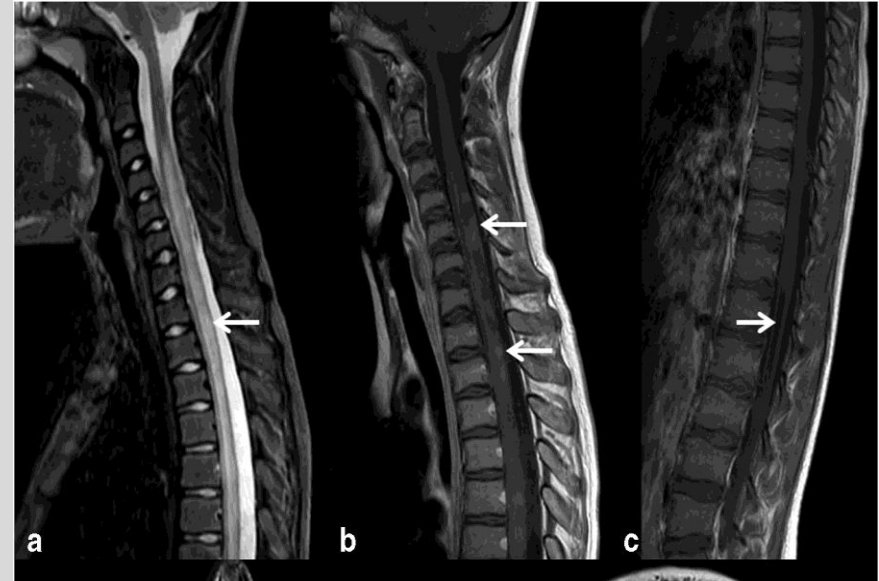
- A 15-year-old boy with anxiety, migraines and ADHD presented with rapidly progressive lower extremity weakness.
- One month prior to the weakness, he developed fever and flu-like symptoms with myalgia in both legs.
- The leg weakness persisted despite resolution of flulike symptoms.

Case 1 continued

- MRI revealed a linear hyperintensity in the gray matter in the thoracic region.
- CSF- predominant lymphocytes

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- **Answer:** ACUTE FLACCID MYELITIS
- Flaccid limb weakness post viral infection
- Decreased muscle stretch reflexes
- Associated with pathogens: enteroviruses D68 and 71, poliovirus and flaviviruses. (Post resp or GI virus).
- More common in children. An outbreak occurs every other year.

- VITAMINS- *Differential Dx of SC lesion*
AKA- Myelopathy
- Vascular
- Infection/post-infectious
- Traumatic
- Autoimmune
- Metabolic /Toxic
- Inflammatory
- Neoplastic
- Structural



DDx for Myelopathy

- V=Spinal cord infarct, vascular malformation
- I=Virus, bacteria, fungus, **post viral**
- T=Decompression illness, trauma
- A=Autoimmune- Lupus, Sjogren's, MS, NMO, MOG
- M=B12 deficiency, drug exposures, nitrous oxide toxicity
- I= Neurosarcoidosis
- N=Spinal cord tumors- primary or metastatic
- S=Syrinx or other congenital anomalies of the cord

Symptoms of Myelopathy

- Depends on the tracts affected and the disease process.
 - Gait instability/imbalance
 - Stiffness or clumsiness
 - Weakness in the arms or legs
 - Urinary urgency /bowel incontinence-
 - Sensory level- sensitive squeezing belt, irritating bandlike sensation at the superior margin and numbness or tingling below the level.
 - Episodic muscle spasms, which cause unintentional extension of legs or cramping in the calves

Signs of a myelopathy

- Flaccid paralysis (corticospinal tract) evolves to
- Weakness with increased tone, brisk DTRs
- Babinski sign
- Hoffman sign (if cervical cord is involved). Fingers and thumb flex involuntarily to a stimulus.

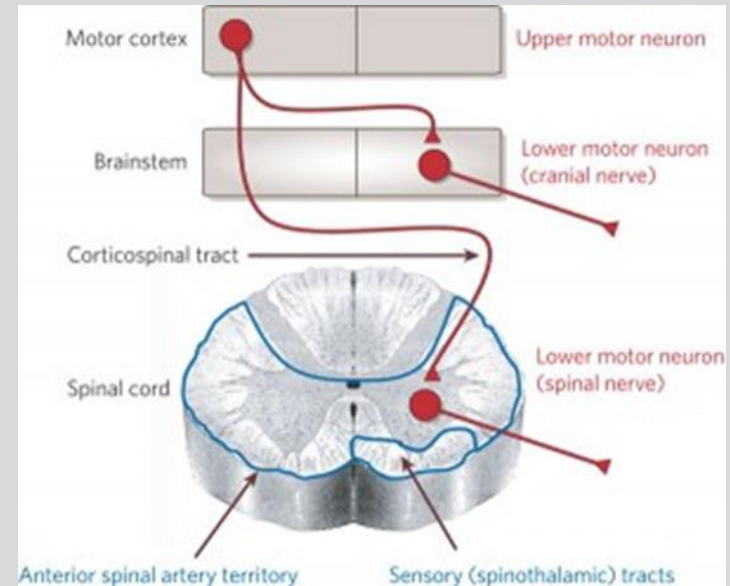
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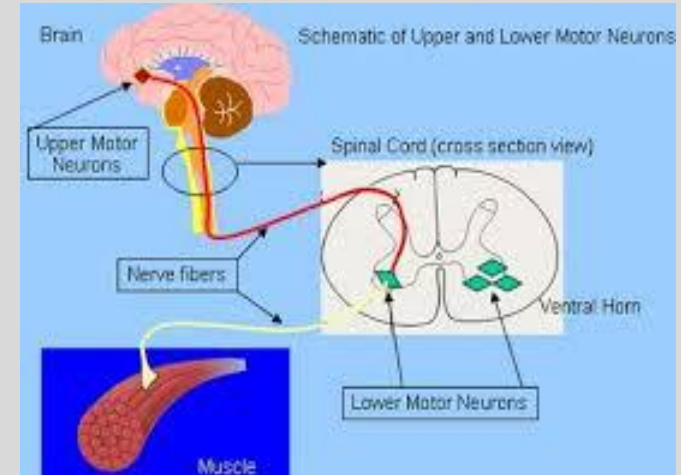
Upper motor neurons

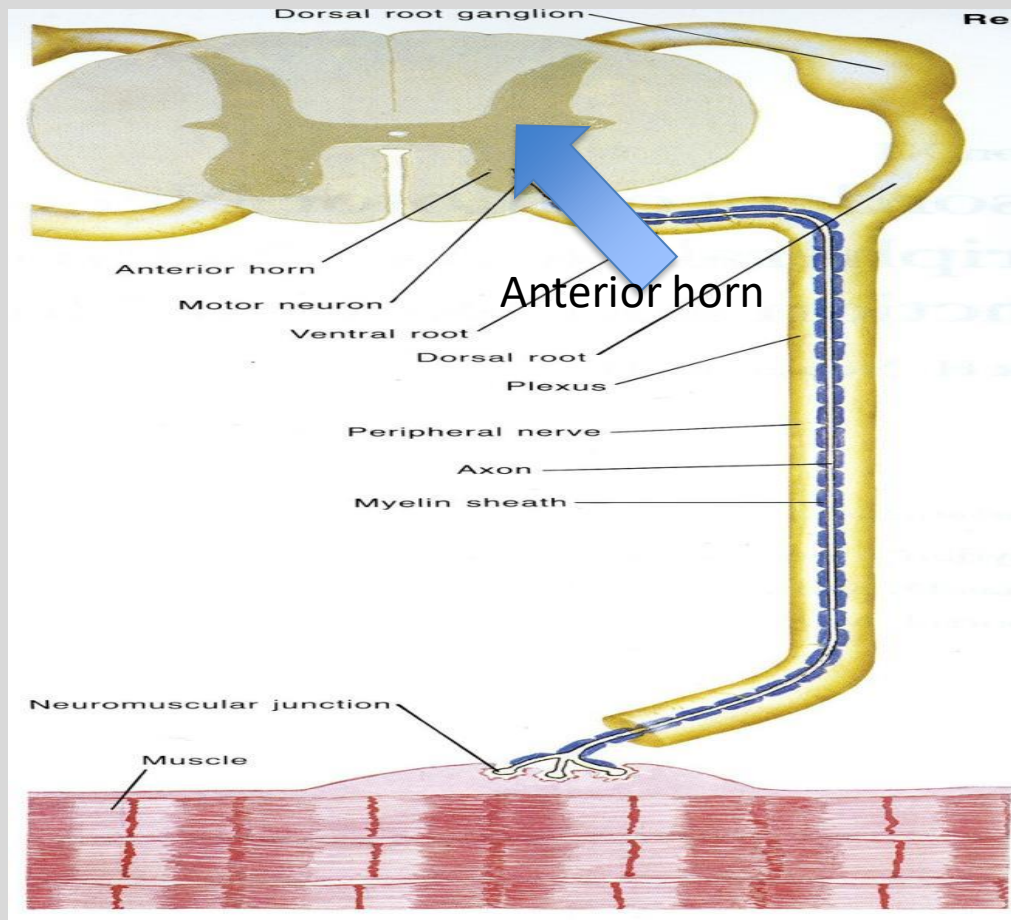
- UMNNs originate in the primary motor cortex of the brain.
- Their axons comprise the **corticobulbar tract** (connecting to the brainstem) and the **corticospinal tract** (connecting to the spinal cord).



Lower motor neurons

- Also referred to as **alpha motor neurons** or **anterior horn cells**.
- Located in motor nuclei in the brainstem or the anterior gray matter of the spinal cord.
- Axons connect to muscles of the bulbar region or limbs.





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Case 2

- The year is 1999. A 73-year-old man from Queens presented to the hospital with fever, headache rash and body aches.
- This progressed to flaccid paralysis.
- Diminished to absent reflexes
- Dysarthria and dysphagia
- Respiratory failure- requiring intubation.



West Nile Virus

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- Neuroinvasive disease is uncommon. Occurs in less than 1% of people.
- 5-10% of neuroinvasive cases are associated with myelitis-specifically a polio-like illness with acute flaccid paralysis, decreased reflexes and bowel and bladder dysfunction. *This is a flavivirus.*
- It often starts with fever, rash and myalgia. More common is meningitis/encephalitis.

Anterior Horn Syndrome

- Refers to lower motor neuron degeneration
- Weakness
- Preservation of all sensory modalities
- Examples:
 - Spinal muscular atrophy
 - Spinal and bulbar muscular atrophy (Kennedy disease)
 - **West Nile myelitis**
 - Poliomyelitis

Spinal Muscular Atrophy

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- The spinal muscular atrophies are a group of inherited disorders that cause degeneration of anterior horn cells.
- Most common type is caused by a variation in the SMN (survival motor neuron 1).
- There are now 9 recognized types identified. All involve varying degrees of weakness
- Clinical presentation varies from early death in infancy (type 0, 1A) to walking independently as an adult (type 4)

Poliomyelitis

- A highly infectious enterovirus.
- 1% of infected patients will develop acute paralytic syndrome due to infection of the anterior horns cells.
- Painful muscle spasms occur before flaccid weakness (spinal and/or bulbar)
- Vaccination remains the main means of polio prevention.
- Iron lung (1940s-early 1950s)/vaccine 1955

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Case 3

- A 67-year-old man presented with an 8-month history of slurred speech.
- Over that same time period he had weakness of the left hand without numbness.
- Cramping was noted in his hands and legs.
- He admitted to crying spontaneously.

Clinical features of ALS

Motor symptoms

- Weakness of limbs with atrophy
- Bulbar (speaking, chewing, swallowing)
- Brisk reflexes

Pseudobulbar affect

- Laughing or crying- not under voluntary control

Cognitive features

- Small proportion may also have frontotemporal dementia

ALS

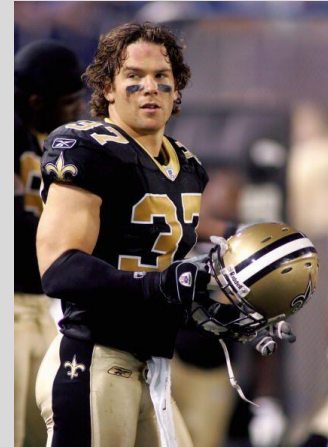
- Combination of upper and lower motor neuron symptoms.
- LMN- **weakness, fasciculations, atrophy**
- UMN-**spasticity, hyperreflexia, and pathologic reflexes** (i.e. Babinski).
 - Note: A retained reflex in a weak limb is considered ***abnormal*** and is a clinical clue.
 - Pelvic floor- urinary and rectal sphincters are unaffected.

Dx ALS

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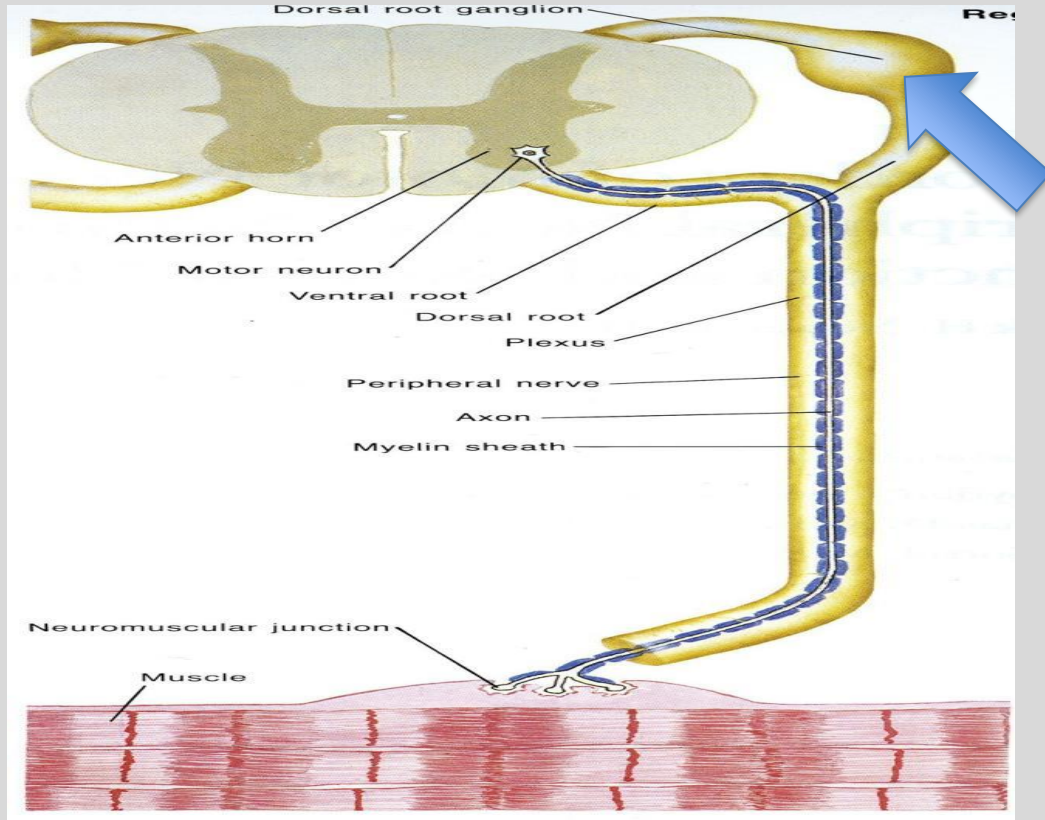
- Combination of history, exam findings, electrophysiologic features. Exclude other causes.



Images: Wikipedia

Continuum- Lifelong Learning in Neurology. Peripheral Nerve and Motor Neuron Disorders. October 2023 VOL. 29 NO. 5

Dorsal Root Ganglion



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Case 4

- 78-year-old man with other comorbidities presents with several days of severe pain on the right side of his thorax.
- On examination there was a vesicular rash present along the T6 distribution.

Herpes Zoster

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http://www.medicinenet.com/_picture_slideshow/article.htm

Varicella-zoster Virus

- Herpes viruses (HSV-1, HSV-2 and VZV) reside in the dorsal root and trigeminal ganglia.
- VZV causes 2 conditions: the primary infection (varicella-chicken pox) and the reactivation infection, (herpes zoster-shingles).
- Zoster is a common complication of a prior VZV infection. Vesicular lesions occur along a dermatome.

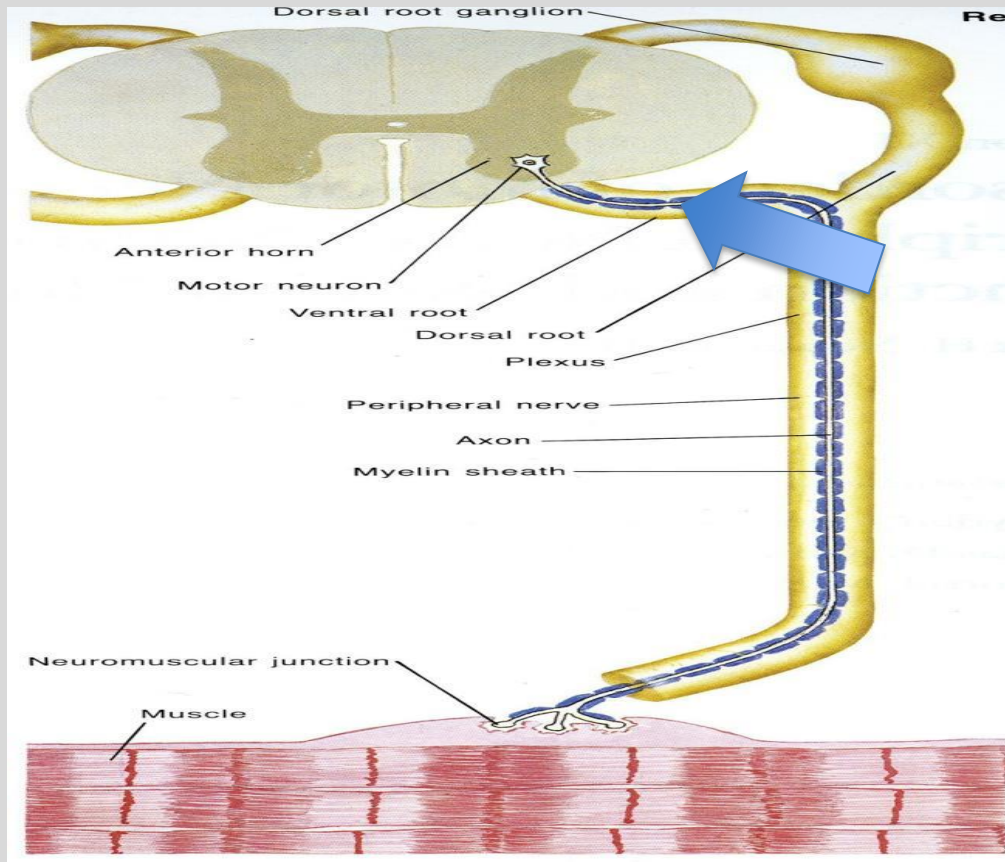
Possible Etiologies of a DRG Lesion- AKA Sensory Neuronopathy

- **Infectious:** HIV, HTLV, EBV, Zika, **VZV**,
- Paraneoplastic: small cell lung ca, breast, ovarian
- Autoimmune: Sjogren's, SLE, Rheumatoid arthritis
- Drug-or drug toxicity: Cisplatin (a chemo agent)
- Hereditary: Fabry (less common)

Features of a DRG lesion

- *Non-length dependent, asymmetric sensory deficits and preserved motor strength*
- Proximal areas may be affected prior to distal areas. This includes upper extremities, face and trunk.
- Pain is common.
- Profound sensory loss with gait and limb ataxia.
- Absent deep tendon reflexes

Ventral Root



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Case 5

- 45-year-old man presents to the ED with severe pain in the left lumbar region.
- Onset was 1 week ago when he sneezed.
- He experienced radiation of pain down the left leg.
- Progressively worsening over the past week.

Case 5 continued

- A foot drop is present (3/5 dorsiflexion weakness) on the left leg.
- Ankle reflex is diminished on the left.
- Sensory loss on the lateral aspect of the left leg.

Nerve Root- DDx

- Most common- intervertebral disk herniation and spondylosis.
- Synovial cysts (less common)
- Infection (osteomyelitis and diskitis)
- Vertebral fracture
- Bony malignant disease
- Epidural hematoma

Symptoms of a Radiculopathy

- Radiating pain
- Reduced reflex along spinal root.
- Sensory loss may be present.
- Muscles weakness supplied by corresponding nerves.

Image of herniated disc

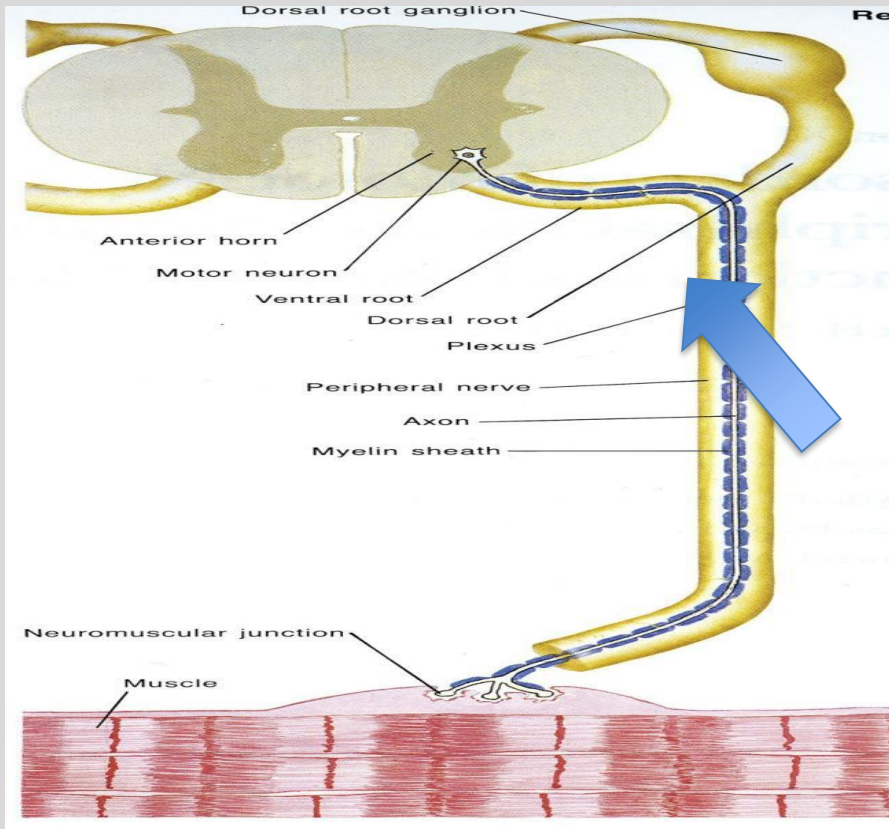
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Mayfield clinic

Plexus



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Case 6

- A 25-year-old man presents with severe attacks of pain in the left upper arm, followed by patchy muscle weakness.
- On exam- there is weakness, patchy sensory loss and atrophy in the upper arm.

Neuralgic Amyotrophy- Parsonage-Turner syndrome (a type of plexopathy)

- Typically presents with acute onset of shoulder pain (unilateral).
- Pain resolves, but then evolves into sensory loss and weakness of the affected limb.
- Risk factors: recent viral illness, surgery
- This is inflammatory in nature.

Brachial Plexus



F. Netter M.D.
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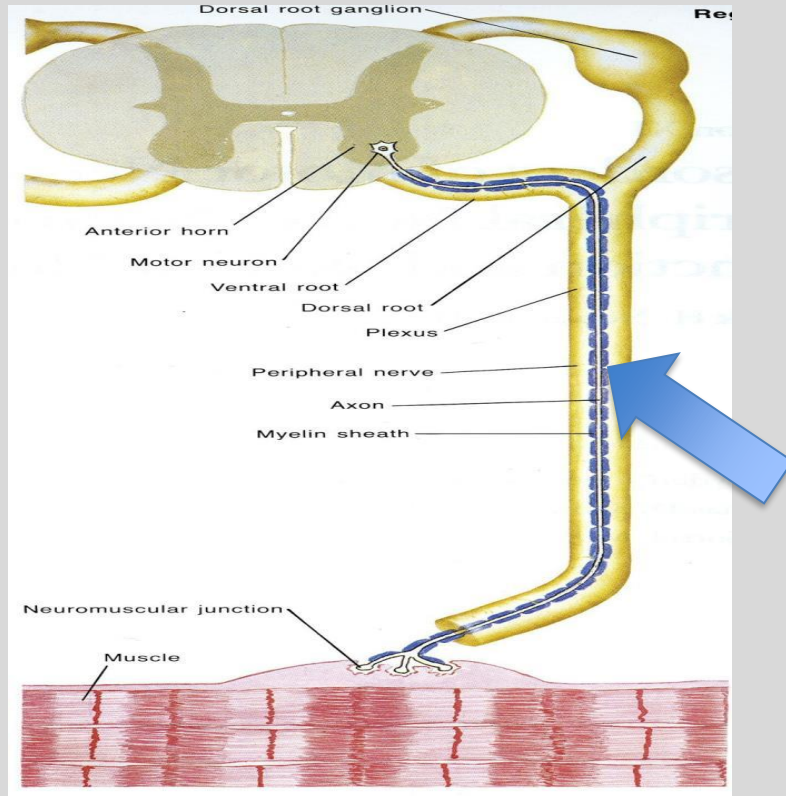
Plexopathy DDx

- Trauma (MVA, birth)
- Neoplasm (primary or metastatic)
- Radiation- occurs months to years later
- Compression/entrapment
- Infection-herpes zoster
- Inflammatory/Immune mediated/ Autoimmune
- Hereditary
- Idiopathic

Peripheral Nerve

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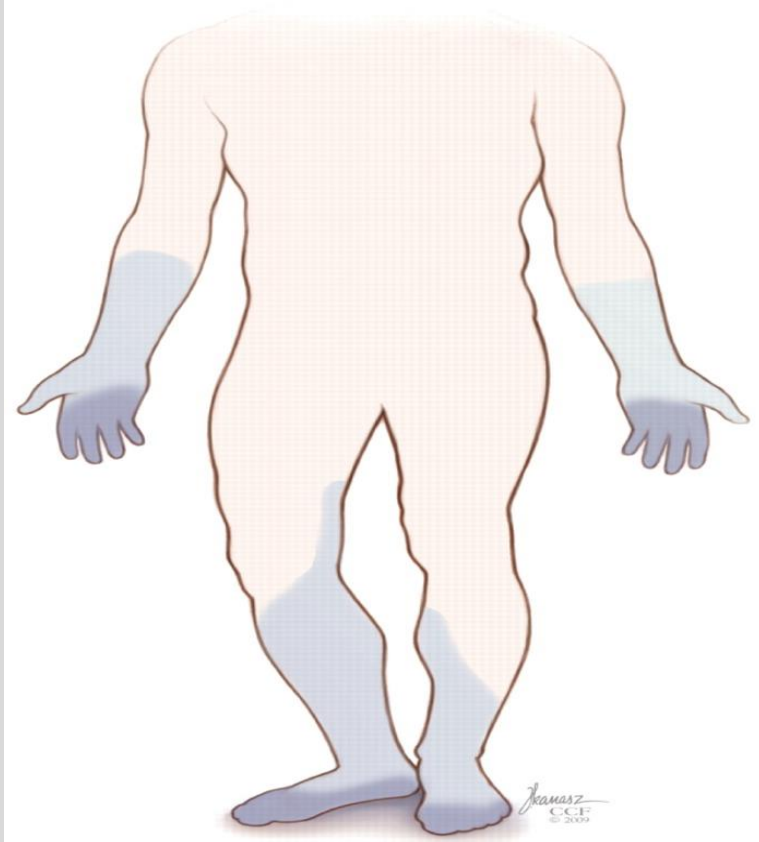


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Case 7

- 56-year-old man with poorly controlled diabetes, HTN and elevated cholesterol presents with cc:
- “My feet feel numb and I’ve been dropping things lately”.
- On examination there is reduced sensation in a stocking glove distribution hands/feet.

Distal symmetric polyneuropathy



Hepatitiscnewdrugs.blogspot.com

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Distal symmetric polyneuropathy

- Diabetes or prediabetes-most common
- Infectious Neuropathies
- Chemotherapy
- Chronic alcohol use and other toxins
- Vitamin B12 deficiency (uncommon)
- Hereditary- hereditary motor sensory neuropathy
- Entrapment (carpal tunnel syndrome)-
mononeuropathy

Leprosy

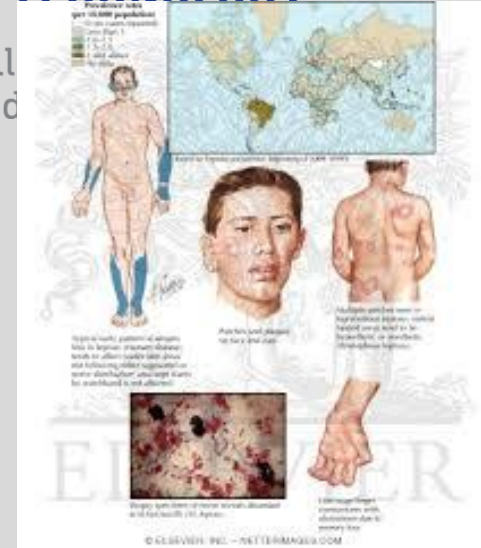
- A common cause of chronic, severe, mutilating infectious neuropathy in tropical and subtropical parts of the world (endemic in Africa, eastern Pacific and western Mediterranean).
- Cause is *Mycobacterium leprae* or *lepromatosis*.
- Some armadillos are naturally affected
- Moloka'i- a former leper colony

Images: pinterest.com and netter plate

Continuum: peripheral nervous system disorders, Oct. 2023 . Vol 20 No. 5,

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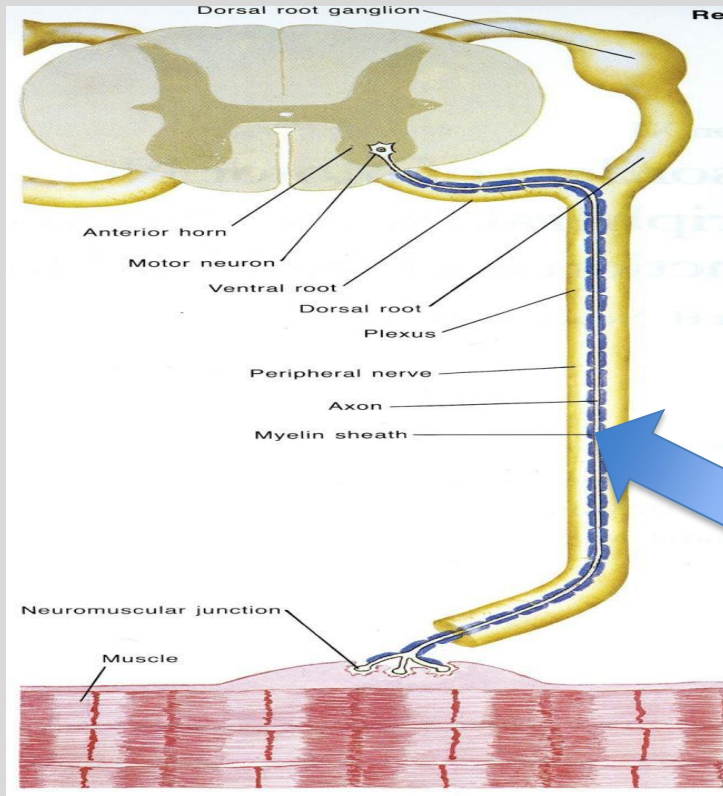
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Myelin Sheath

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Myelin sheath

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Case 8

- A-25-year-old woman presents with back pain and numbness in her feet.
- Over the past day, numbness has been spreading up to her knees.
- She had a gastroenteritis a few weeks ago.
- **Reflexes are absent**
- **Lumbar puncture- albuminocytologic dissociation (elevated protein with a normal cell count).**

Guillain-Barre Syndrome

- Encompasses a group of related disorders:

Acute inflammatory demyelinating polyradiculoneuropathy (AIDP) is the most common form.

- Post-infectious and paralytic
- CMV, Mycoplasma pneumonia, EBV, influenza A, H. flu.
- Campylobacter jejuni (causes GI) is most frequently associated bacteria.
- GBS has been found to occur after Zika virus and SARS COVID-19 infections.

Why ICU care is necessary

- Dysautonomia
- Bulbar dysfunction
- Severe or rapidly worsening weakness
- Evolving respiratory distress (inability to count 15 or more during a single breath)
- Rapid decline in respiratory function (>30% in 24 hours)

Treatment of GBS

- Supportive care is the cornerstone of treatment. Evidence based research has shown that
- **IVIG (IV immunoglobulin)** and
- **Plasma Exchange** are equally efficacious.
- Either one should be started ASAP once dx is confirmed.
- Steroids are **not** effective

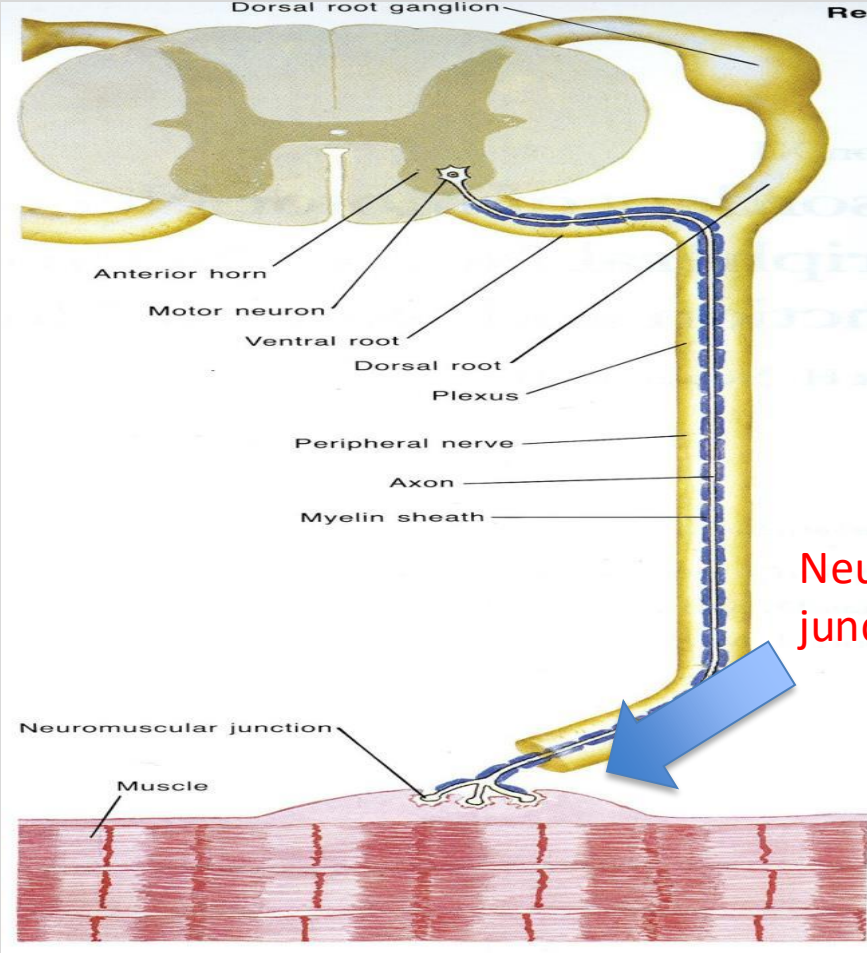
CIDP

- Symmetric proximal and distal weakness and sensory loss develop over at least **2 months** in a progressive, stepwise or relapsing course.
- Reduced or absent reflexes.
- Usually no antecedent illness.
- Less common dysautonomia and respiratory failure
- Frequent postural tremor

Treatment for CIDP

- Treated with one of 3 first line therapies:
 - Immunoglobulins- IV or SQ
 - Corticosteroids
 - Plasma Exchange

50-70% respond to one of these therapies



Neuromuscular
junction

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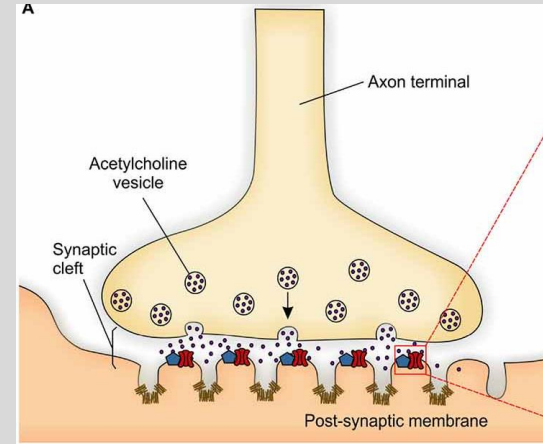
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Case 9

- A 23-year-old healthy woman presents with several weeks of intermittent diplopia and difficulty with climbing stairs. It is worse towards the end of the day.
- On exam- difficulty sustaining upgaze, mild lid ptosis and bilateral hip flexor weakness.

DDx of NMJ Disorders

- Myasthenia Gravis
- Lambert Eaton Myasthenic Syndrome
- Botulism
- Immune Checkpoint Inhibitor-related MG



Ptosis (drooping of the eyelid)



 ADAM.

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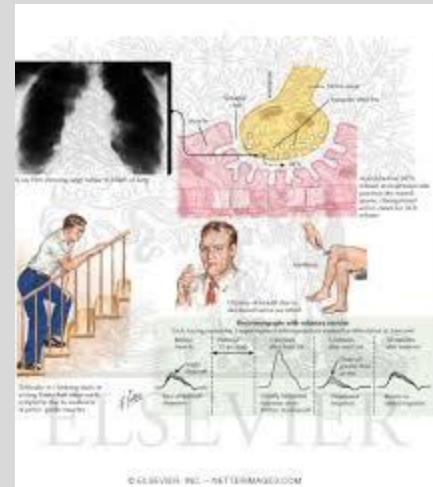
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Autoimmune Myasthenia Gravis

- Autoimmune disease- causing fatiguable ocular, limb and bulbar muscle weakness-to different degrees.
- Peaks in younger women and older men.
- Antibodies against AChR or MuSK in the majority of patients.
- Highly treatable with immunomodulation.
- Thymectomy may be warranted.

Lambert Eaton Myasthenic Syndrome

- Rare autoimmune disease that affects **presynaptic** neuromuscular transmission.
- Antibodies against voltage gated calcium channels.
- Half of patients with small cell lung cancer.
- Proximal muscle weakness, hyporeflexia/areflexia and dysautonomia.
- Weakness and DTRs briefly improves after exercise.
- Treatment: 3,4-diaminopyridine/ oncologist for CA

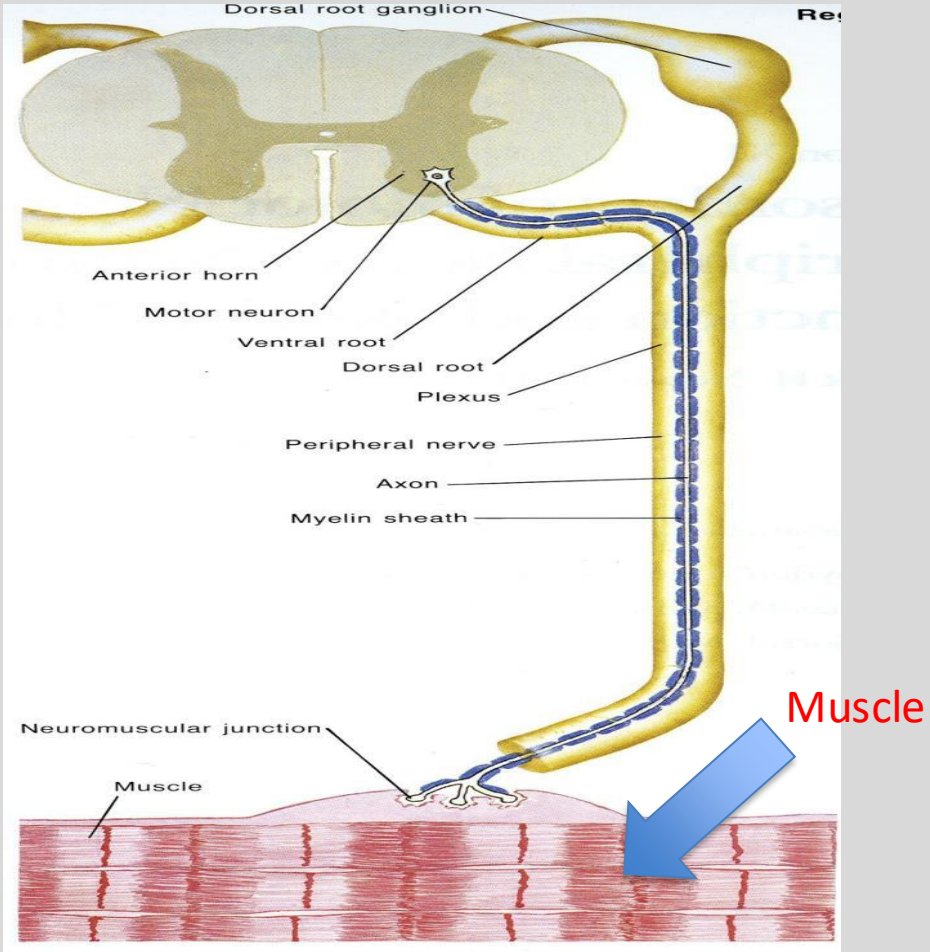


Botulism

- A paralytic illness caused by Botulinum neurotoxins, *produced by a strain of Clostridium botulinum:inhibits ACh release at the neuromuscular junction.*
- Acute descending flaccid paralysis.
- Enters the body via: ingestion (foodborne), wound colonization, intestinal colonization (infant and adult), cosmetic or aerosolized.
- Blurred vision, slurred speech, descending paralysis, resp failure.

Honey may
quiet them,
but botulism
can kill them





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Case 10

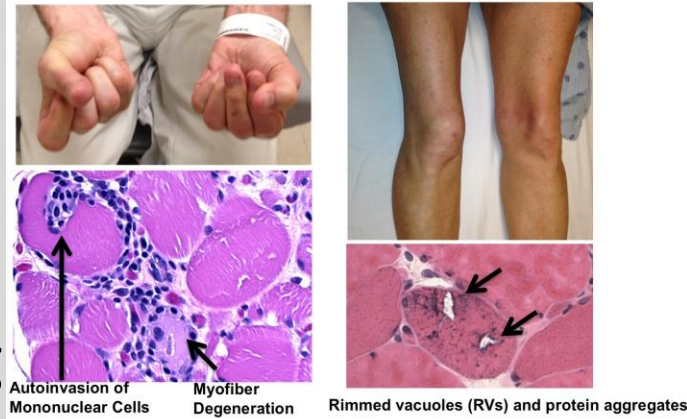
- A 70-year-old man presents with a 2-year history of progressive lower limb weakness. “Knees give out”.
- Weakness of deep finger flexors, and left quadriceps in an asymmetric pattern.

Inclusion Body Myositis

- The most common myopathy in adults.
- Affects males twice as often as females
- Knee extensor weakness is most prominent with loss of ambulation over 10 years.
- Over time there is loss of swallowing and respiratory involvement as cause of death.

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Symptoms seen in a Myopathy

- Exercise intolerance
- Muscle atrophy
- Weakness
- Muscle Cramps
- Contractures
- Muscle hypertrophy (hereditary types)
- Myoglobinuria
- Pain in muscles at rest or with exertion (depends on type)

Differential of Muscle Disorders

- Inherited myopathies
 - Dystrophinopathies
- Inflammatory Myopathies
 - Inclusion body
 - Polymyositis
- Acquired myopathies
 - Immune checkpoint inhibitor
 - Statin induced
 - Thyroid ophthalmopathy

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Muscular Dystrophy

- Myotonic dystrophy
- Duchenne muscular dystrophy
- Becker muscular dystrophy
- Facioscapulohumeral, Emery-Dreifuss MD
- Limb-girdle muscular dystrophy

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Summary Slide

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- Summary Point 1: Localize the lesion based on clinical presentation and generate a differential diagnosis based on the localization.
- Summary Point 2: Do not miss a treatable diagnosis and know what the appropriate treatment is.
- Core References:
 - Continuum- Lifelong Learning in Neurology. Muscle and Neuromuscular Junction Disorders. October 2025 VOL. 31 NO. 5
 - Continuum- Lifelong Learning in Neurology. Peripheral Nerve and Motor Neuron Disorders. October 2023 VOL. 29 NO. 5.
 - Continuum- Lifelong Learning in Neurology. Spinal Cord Disorders. February 2024 Volume 30 No.1

Lecture and Lab Feedback Form:

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